

Seong-Jun Kang, Ph.D.

Experimental and computational biomedical scientist: multiplexed molecular imaging, single-cell and spatial biology, quantitative image analysis

Postdoctoral Associate applicant · Bathe Lab, MIT Department of Biological Engineering

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RESEARCH PROFILE

I am an immunologist working with primary mammalian cells, multiplexed protein and RNA measurements, and quantitative imaging. My hands-on experience includes end-to-end GeoMx DSP, MACSima cyclic immunofluorescence, confocal microscopy, and automated single-cell library preparation. I would apply donor-aware analysis and held-out evaluation to PRISM assay validation, imaging automation, and drug-response studies in human iPSC-derived neurons.

■ POSITION-RELEVANT EXPERTISE

- **Multi-modal cellular measurement.** 10× single-cell workflows, end-to-end GeoMx DSP, Visium spatial transcriptomics, MACSima cyclic multiplexed immunofluorescence, high-parameter flow cytometry, and targeted proteomics.
- **Computational analysis.** Sequencing-data analysis in Python and R using statistical and machine-learning tools, multimodal integration, and donor- or case-level evaluation; co-authored an AIDD workflow spanning target discovery, binding prediction, virtual screening, and lead optimization.
- **Wet-lab and translational experiments.** Primary-cell culture and DNA/RNA processing; 10× Chromium scRNA-seq and GeoMx DSP spatial workflows; SPR and ELISA binding assays; and Olink proteomics.
- **Animal experiments.** EAE and Alzheimer's disease models; skin- and islet-transplantation studies; animal breeding and longitudinal immune monitoring.
- **Program and study execution.** Multi-site sample QC, non-human-primate studies, and proposal writing and execution for approximately \$1.3 million in industry awards.

■ EDUCATION

Ph.D.	Seoul National University, Seoul, Republic of Korea
2014-2023	Department of Biomedical Sciences
	<i>Thesis: Exploration of the Role of Regulatory T cells in Various Pathogenic Contexts</i>
	Advisor: Chung-Gyu Park, M.D., Ph.D.

M.S. 2012–2014	Seoul National University , Seoul, Republic of Korea Department of Biomedical Sciences <i>Thesis: Enhanced LPS-Induced Monocyte Activation by Recombinant Human Soluble CD14</i> Advisor: Chung-Gyu Park, M.D., Ph.D.
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B.S. 2008–2012	Sungkyunkwan University , Suwon, Republic of Korea Department of Genetic Engineering
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■ RESEARCH & PROFESSIONAL EXPERIENCE

Researcher (postdoctoral) 2025–present	Yonsei University Wonju College of Medicine · Dept. of Physiology / Organelle Medicine Research Center, Wonju, Republic of Korea Contributed to a project examining associations between skin immunity and Alzheimer's disease using public transcriptomic and clinical datasets.
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Associate Director Sep 2023–Aug 2025	PB Immune Therapeutics Inc. · Dept. of Basic Research, Seoul, Republic of Korea Led anti-CD40 candidate evaluation, including primate studies, external-study coordination, advancement decisions, and single- cell/spatial workflow development.
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Manager May 2021–Aug 2023	PB Immune Therapeutics Inc. · Dept. of Basic Research, Seoul, Republic of Korea Led KDDF-funded anti-CD40 lead-selection work and wrote proposals for multiple-sclerosis and pemphigus-vulgaris programs.
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Researcher Sep 2019–Apr 2021	Xenotransplantation Research Center, Seoul National University , Seoul, Republic of Korea Rhesus tissue processing and transplant immune monitoring.
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Military Service Sep 2016–Aug 2019	Research Personnel, Transplantation Research Institute, Seoul National University MRC , Seoul, Republic of Korea
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Teaching Assistant Sep 2014–Aug 2016	Seoul National University College of Medicine · Dept. of Microbiology and Immunology, Seoul, Republic of Korea
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BK21 Research Student Mar 2012–Aug 2014	BK21 Plus Biomedical Science Project, Seoul National University , Seoul, Republic of Korea
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PUBLICATIONS

20

TOTAL

17

PUBLISHED

1

IN REVISION

2

IN PREPARATION

6

FIRST / CO-FIRST
Published & in revision

 [Google Scholar · Full Profile, Citations & h-index ↗](#)

FIRST / CO-FIRST AUTHOR

- 2026 Woogil Song*, **Seong-Jun Kang***, Taehee Stella Kim*, Seyoung Jung, Sung Hee Kim, Su Chan Park, Sung Hae Chang, Eun Young Lee, Tae-Gyun Kim, Hyun Je Kim, Jeong Seok Lee. "Conserved PDE4D short-isoform bias defines pathogenic Th1.17 state across human inflammatory diseases." *Nature Immunology*. [in revision](#)
- 2025 **Seong-Jun Kang***, Yong-Hee Kim*, Thuy Nguyen-Phuong*, Yijoon Kim, Jin-Mi Oh, Jae-chun Go, DaeSik Kim, Chung-Gyu Park, Hyunsu Lee, Hyun Je Kim. "Immune cell-enriched single-cell RNA sequencing unveils the interplay between infiltrated CD8⁺ T resident memory cells and choroid plexus epithelial cells in Alzheimer's disease." *Journal of Neuroimmunology*. [Download PDF](#)
- 2025 **Seong-Jun Kang***, Jeong-Ryeol Gong*, Seon-Pil Jin, Jin-Mi Oh, Hyunjin Jin, Yuji Lee, Yewon Moon, Dongjun Kim, Hyo Jeong Nam, Hyun Seung Choi, Sanha Hwang, Yun Jung Huh, Kyung Yeon Han, Jihwan Moon, Jongsuk Chung, Woong-Yang Park, Chung-Gyu Park, Hyun Je Kim, Jeong Eun Kim. "Deciphering Dysfunctional Regulatory T Cells in Atopic Dermatitis." *Allergy*. [Download PDF](#)
Co-first author. Single-cell and functional dissection of skin-homing regulatory T cells in atopic dermatitis, with OX40 up-regulated alongside impaired suppressive function.
- 2024 Jong-Min Kim*, **Seong-Jun Kang***, So-Hee Hong*, Hyunwoo Chung, Jun-Seop Shin, Byoung-Hoon Min, Hyun Je Kim, Jongwon Ha, Chung-Gyu Park. "Long-term control of diabetes by tofacitinib-based immunosuppressive regimen after allo-islet transplantation in diabetic rhesus monkeys that rejected previously transplanted porcine islets." *Xenotransplantation*. [Download PDF](#)
Co-first author. Long-term diabetes control with a tofacitinib-based regimen after allo-islet transplantation in rhesus macaques, with longitudinal immune monitoring.
- 2022 Youngkyoung Lim*, **Seong-Jun Kang***, Beom Keun Cho, Hyun Je Kim, ... Chong Hyun Won, Chung-Gyu Park. "Intra-tumoral heterogeneity and immune escape of melanoma arising from congenital melanocytic nevus revealed by spatial gene expression profiling." *JEADV*. [Download PDF](#)

- 2019 Jong-Min Kim*, Jun-Seop Shin*, Byoung-Hoon Min*, **Seong-Jun Kang***, Il-Hee Yoon, Hyunwoo Chung, Jiyeon Kim, Eung-Soo Hwang, Jongwon Ha, Chung-Gyu Park. "JAK3 inhibitor-based immunosuppression in allogeneic islet transplantation in cynomolgus monkeys." *Islets*. [Download PDF](#)
- Co-first author, one of four authors contributing equally. JAK3-inhibitor-based immunosuppression in allogeneic islet transplantation in cynomolgus macaques.

CONTRIBUTING AUTHOR

- 2026 Jung Ho Lee*, Sung Ha Lim*, Hyungdon Kook, **Seong-Jun Kang**, Geon Lee, Brian Hyohyoung Lee, Hyunsung Nam, YongJun Kim, Christine Suh-Yun Joh, Soyoung Jeong, Dongryeol Shin, Hyun Je Kim, Jiyoung Ahn. "Mycosis Fungoides-like Atopic Dermatitis Represents a Th22-dominant Inflammatory Endotype." *Allergy*.
- 2024 Brian Hyohyoung Lee*, Yoon Ji Bang*, Sung Ha Lim*, **Seong-Jun Kang**, Sung Hee Kim, Seunghee Kim-Schulze, Chung-Gyu Park, Hyun Je Kim, Tae-Gyun Kim. "High-dimensional profiling of regulatory T cells in psoriasis reveals an impaired skin-trafficking property." *eBioMedicine*. [Download PDF](#)
- 2024 Youngkyoung Lim*, Beom Keun Cho*, **Seong-Jun Kang**, Soyoung Jeong, Hyun Je Kim, Jiyeon Baek, Ji Hwan Moon, Cheol Lee, Chan-Sik Park, Je-Ho Mun, Chong Hyun Won, Chung-Gyu Park. "Spatial transcriptomic analysis of tumour-immune cell interactions in melanoma arising from congenital melanocytic nevus." *JEADV*. [Download PDF](#)
- 2023 Sunyoung Jung*, Sunho Lee, Hyun Je Kim, Sueon Kim, Ji Hwan Moon, Hyunwoo Chung, **Seong-Jun Kang**, Chung-Gyu Park. "Mesenchymal stem cell-derived extracellular vesicles subvert Th17 cells by destabilizing ROR γ t through posttranslational modification." *Experimental & Molecular Medicine*.
- 2022 Sangho Lee, Han-Teo Lee, Young Ah Kim, Il-Hwan Lee, **Seong-Jun Kang**, Kyeongpyo Sim, Chung-Gyu Park, Kyungho Choi, Hong-Duk Youn. "The optimized core peptide derived from CABIN1 efficiently inhibits calcineurin-mediated T-cell activation." *Experimental & Molecular Medicine*.
- 2021 So Hee Hong, Hyun Je Kim, **Seong-Jun Kang**, Chung-Gyu Park. "Novel Immunomodulatory Approaches for Porcine Islet Xenotransplantation." *Current Diabetes Reports*. [Download PDF](#)
- 2020 Sunho Lee, Sueon Kim, Hyunwoo Chung, Ji Hwan Moon, **Seong-Jun Kang**, Chung-Gyu Park. "Mesenchymal stem cell-derived exosomes suppress proliferation of T cells by inducing cell cycle arrest through p27kip1/Cdk2 signaling." *Immunology Letters*.

- 2019 Hyun Je Kim, Ji Hwan Moon, Hyunwoo Chung, Jun-Seop Shin, Bongji Kim, Jong Min Kim, Jung-Sik Kim, Il-Hee Yoon, Byoung-Hoon Min, **Seong-Jun Kang**, Yong-Hee Kim, Kyuri Jo, Joungmin Choi, Heejoon Chae, Won Woo Lee, Sun Kim, Chung-Gyu Park. "Bioinformatic analysis of peripheral blood RNA-sequencing sensitively detects the cause of late graft loss following overt hyperglycemia in pig-to-non-human primate islet xenotransplantation." *Scientific Reports*.
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- 2018 Byoung-Hoon Min, Jun-Seop Shin, Jong-Min Kim, **Seong-Jun Kang**, Hyun-Je Kim, Il-Hee Yoon, Su-Kyoung Park, Ji-won Choi, Min-Suk Lee, Chung-Gyu Park. "Delayed revascularization of islets after transplantation by IL-6 blockade in pig-to-non-human primate islet xenotransplantation model." *Xenotransplantation*.
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- 2018 Jun-Seop Shin, Jong-Min Kim, Byoung-Hoon Min, Il Hee Yoon, Hyun Je Kim, Jung-Sik Kim, Yong-Hee Kim, **Seong-Jun Kang**, Jiyeon Kim, Hee-Jung Kang, Dong-Gyun Lim, Eung-Soo Hwang, Jongwon Ha, Sang-Joon Kim, Wan Beom Park, Chung-Gyu Park. "Pre-clinical results in pig-to-non-human primate islet xenotransplantation using anti-CD40 antibody (2C10R4)-based immunosuppression." *Xenotransplantation*.
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- 2018 Jong-Min Kim, Jun-Seop Shin, Byoung-Hoon Min, Il Hee Yoon, **Seong-Jun Kang**, Won-Young Jeong, Sang-Joon Kim, Chung-Gyu Park. "Pre-Clinical Results of Islet Allo-Transplantation Using JAK Inhibitor as Replacement for Tacrolimus in Cynomolgus Monkeys." *Transplantation*.
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- 2018 Jung-Sik Kim, Hyunwoo Chung, Nari Byun, **Seong-Jun Kang**, Sunho Lee, Jun-Seop Shin, Chung-Gyu Park. "Construction of EMSC-islet co-localizing composites for xenogeneic porcine islet transplantation." *Biochemical and Biophysical Research Communications*.

MANUSCRIPTS IN PREPARATION

- 2026 Kangsan Roh*, Sabyasachi Das*, **Seong-Jun Kang**, Olga Barreiro, Rodrigo Javier Gonzalez, Timothy P. Padera, Shraddha Shirguppe, Ahlim Lee, Youngju Song, Pazhanichamy Kalailingam, Claire E. Fong, Aarushi Gandhi, Meehyun Kim, Diana Tambala, Adam Vernon Crain, Casey A. Maguire, Ulrich H. Von Andrian, Christian L. Lino Cardenas, Sekeun Kim, Benjamin P. Kleinstiver, Mark E. Lindsay, Patricia L. Musolino, Rajeev Malhotra. "In-vivo base editing of ACTA2 R179H restores lymphatic function and immune cell homeostasis in a murine model of Multisystemic Smooth Muscle Dysfunction Syndrome (MSMDS)." **in preparation**

Co-author (immunology). Ran the Olink plasma proteomics across 12 patients and 6 controls and the flow-cytometry immunophenotyping of lymph-node leukocytes.

2026 Sung Ha Lim*, Jiyeon Oh*, Kyu-Hee Hwang, Taesic Lee, **Seong-Jun Kang**, Eung Ho Choi, Seung-Kuy Cha. "Skin-brain transcriptomic convergence: an integrative framework for predicting Alzheimer's disease from cutaneous aging signatures." **in preparation**

Co-author (methodology, single-cell analysis). Built the method that turns a meta-analysis of 8 skin-aging transcriptome datasets into gene sets and uses them to classify Alzheimer's disease in human brain transcriptomes. Reanalyzed 4 public skin and brain single-cell cohorts (295,243 cells and nuclei) with donor-level pseudobulk to map the candidate genes to specific cell types.

■ TECHNICAL EXPERTISE

Multiplexed in situ detection and imaging	GeoMx DSP run end to end — slide preparation, oligonucleotide-barcoded probe hybridisation, immunofluorescence staining, fluorescence-guided ROI selection, instrument operation, UV cleavage and barcode collection. MACSima cyclic multiplexed immunofluorescence, iterative stain-image-erase rounds on one section. Olink targeted plasma proteomics; Visium spatial transcriptomics. High-parameter fluorescence panel design up to 24 markers. Spinning-disk confocal (Nikon CSU-W1 SORA) with quantitative image analysis.
High-content image and data analysis	Segmentation, feature extraction and model scoring on public iPSC-neuron imaging screens. Batch- and donor-aware modelling, held-out evaluation, calibrated confidence. Automated library preparation on 10x Chromium Connect .
Single-cell / spatial / multi-omic	scRNA-seq, scTCR-seq, snRNA/snATAC; 10x 5'/3' library generation and 10x Chromium Connect automated preparation. Single-cell sequencing data analysis and interpretation in Python and R. Spatial: GeoMx DSP run end to end, Visium , MACSima cyclic multiplexed immunofluorescence.
Computational	Python, R, Linux, and Git ; scikit-learn, XGBoost/LightGBM/CatBoost, and PyTorch. Donor- or case-level validation and multimodal analysis of imaging, transcriptomic, and clinical data.
Cell, tissue & immunological assays	Primary human T-cell isolation, expansion and functional assays; monocyte and dendritic-cell differentiation; mesenchymal stem cell culture. Enzymatic tissue digestion, Percoll enrichment, cell and nucleus isolation. High-dimensional flow cytometry (panels up to 24 markers); multiplex panel design; 10-color 6-way FACS. Olink targeted plasma proteomics (immune & neurology panels).

Microscopy & animal models

Spinning-disk confocal (Nikon CSU-W1 SORA), point-scanning confocal (Leica SP8), two-photon; IHC/IF (frozen & FFPE) with quantitative image analysis. Non-human primates across three species (rhesus, cynomolgus, marmoset), including EAE and islet-transplantation studies; mouse immune profiling.

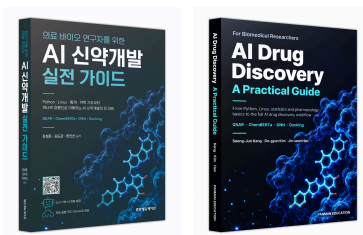
Study execution & collaboration

Study design, external-study coordination, multi-site sample QC, milestone tracking, regulatory documentation, and data review.

BOOKS

Seong-Jun Kang, Do-gyun Kim, Jin-seon Han. **AI Drug Discovery: A Practical Guide for Biomedical Researchers** (title translated). PanmunEducation. September 21, 2026. 548 pages. ISBN 979-11-5943-583-6. [YES24 Global](#)

The book covers molecular and protein data preparation, QSAR, ChemBERTa, graph neural networks, docking, virtual screening, candidate optimization, and follow-up experimental validation.



PATENTS

2024 **Antagonistic Anti-CD40 Humanized Antibody.** Republic of Korea, Appl. No. 10-2024-0085972; registered 2024-07-01. Co-inventor.

2019 **Immunosuppression Composition Comprising JAK Inhibitor.** Republic of Korea, Appl. No. 10-2014-3490000; registered 2019-08-20.

AWARDS & SCHOLARSHIPS

AWARDS

Jun 2024 **Grand Prize**, 2024 GPTers AI Hackathon (Project GAIA Team)

Dec 2023 **Grand Prize**, SBA Smart Workathon for Work Process Innovation

Dec 2021 **Excellence Award (Poster Presentation)**, 2021 Research Safety Contest, Seoul National University College of Medicine

Dec 2020 **Excellence Award (Poster Presentation)**, 2020 Research Safety Contest, Seoul National University College of Medicine

Dec 2017 **Best Research Award**, 2017 Infection and Immunity Research Festival,
Department of Microbiology and Immunology, Seoul National University
College of Medicine

Nov 2017 **Best Presentation Award (Poster Presentation)**, 2017 Annual Meeting of
the Korean Association of Immunologists

May 2017 **Best Poster in Congress Award**, 2017 American Transplant Congress,
Chicago

SCHOLARSHIPS

Sep 2019–Aug 2020 **Research Scholarship for Ph.D. Candidates**, National Research
Foundation of Korea (NRF)

Sep 2014–Aug 2016 **Lecture and Research Support Scholarship**, Dean of Seoul National
University College of Medicine

Mar 2012–Aug 2014 **BK21 Research Scholarship**, BK21 Plus Biomedical Science Project, Seoul
National University

SELECTED PRESENTATIONS

Poster KAI Annual Meeting, Incheon (2022) · Cytokine, Hawaii (2022) · American
Transplant Congress, Chicago (2017) · The Transplantation Society, Hong
Kong (2016) · IPITA/IXA/CTS, Melbourne (2015) · TREG, Shanghai (2012)

■ SELECTED GRANT WRITING & PROGRAM FUNDING

INDUSTRY · PB IMMUNE THERAPEUTICS (MANAGER → ASSOCIATE DIRECTOR)

FUNDING

AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
Korea Drug Development Fund (KDDF)	Anti-CD40 antibody lead selection and optimization (multiple sclerosis)	2021– 2023	~\$571,000	Lead proposal writer (>90% of proposal & execution)
Korea Drug Development Fund (KDDF)	Anti-CD40 lead compound (PB101): efficacy and pharmacodynamic biomarkers (pemphigus vulgaris)	2022– 2024	~\$321,000	Lead proposal writer (>90% of proposal & execution)
Ministry of SMEs and Startups (MSS)	Scale-up Funding for Tech Commercialization	2022– 2023	~\$222,000	Lead proposal writer; managed >90% of execution

FUNDING AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
Ministry of SMEs and Startups (MSS)	Early-stage startup R&D: anti-CD40 biologic (PB-40N) to prevent anti-drug antibodies	2021–2022	~\$74,000	Lead proposal writer & core researcher
NAVER Corporation	Engineered regulatory T cells for immune-mediated diseases (PB20X_Treg)	2023–2026	~\$128,000	Proposal & study design (~50%)

DOCTORAL TRAINING · SEOUL NATIONAL UNIVERSITY

FUNDING AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
Ministry of Science and ICT (MSIT)	Multi-layered immune-disease single-cell map for molecular target discovery (SCAID, A9–22–1–05)	2022–2028 (3+3)	~\$1.6M	Participated in proposal writing (>30%)
Ministry of Education (MOE) & Ministry of Science and ICT (MSIT)	Laboratory–Specialized Tech–Transfer & R&BD Grant	2020–2021	~\$40,000	Grant writing, experimental design & technical validation

All grants were held by the institutional Principal Investigator (Ph.D. advisor at SNU; company CEO/CSO at PB Immune Therapeutics); the listed roles reflect the applicant's contribution to proposal writing and project execution, **not as PI**. Budgets converted at ≈ ₩1,400 / USD.

MENTORING AND RESEARCH TRAINING

YEAR	MENTEE / ROLE AT TIME	CURRENT AFFILIATION	RESEARCH TRAINING
2013	Sung Ha Lim, M.D., Ph.D. Student intern	Postdoctoral Fellow; Laboratory of Shadmehr Demehri, M.D., Ph.D.; Department of Dermatology, Massachusetts General Hospital; Harvard Medical School, Boston, MA, USA	Cell culture, Western blotting, flow cytometry, and C57BL/6 mouse breeding.
2014	Soyoung Jeong, Ph.D. Student intern	Postdoctoral Associate; Medical Research Center, Seoul National University, Seoul, Republic of Korea	Cell culture, flow cytometry, and in vivo mouse experiments.
2016	Sowon Choi, M.S. Incoming graduate student	Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea	Mouse pancreatic islet isolation, pancreatic islet beta-cell functional assays, and ion channel assays.

YEAR	MENTEE / ROLE AT TIME	CURRENT AFFILIATION	RESEARCH TRAINING
2021	Hyo Jeong Nam, M.S. Incoming graduate student	Associate Researcher, Human Immune Monitoring Center, Icahn School of Medicine at Mount Sinai, New York, NY, USA	GeoMx DSP, T-cell functional assays, and single-cell RNA sequencing (scRNA-seq) data analysis using R.
2021	Beom Keun Cho, M.S. Incoming graduate student	Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea	GeoMx DSP, mouse breeding, cell culture, T-cell functional assays, and scRNA-seq data analysis using R.
2021	Minji Kang, Ph.D. Incoming graduate student	Postdoctoral Associate; Transplantation Research Institute; Seoul National University Medical Research Center, Seoul, Republic of Korea	GeoMx DSP spatial transcriptomics data analysis and T-cell functional assays.
2021	Jung Ho Lee, M.S. Incoming graduate student	Visiting Scholar; Laboratory of Ulrich von Andrian, M.D., Ph.D.; Department of Immunology, Harvard Medical School, Boston, MA, USA; Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea	scRNA-seq data analysis using R.
2021	Hyun Seung Choi, M.S. Incoming graduate student	Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea	scRNA-seq data analysis using R.
2022	Yuji Lee, M.S. Incoming graduate student	Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea	Regulatory T-cell suppression assays and scRNA-seq data analysis using R and machine-learning models.
2022	Sanha Hwang, M.S. Incoming graduate student	Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea	Regulatory T-cell suppression assays and scRNA-seq data analysis using R and machine-learning models.
2022	Dongjun Kim, M.S. Incoming graduate student	Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea	Regulatory T-cell suppression assays and scRNA-seq data analysis using R and machine-learning models.

YEAR	MENTEE / ROLE AT TIME	CURRENT AFFILIATION	RESEARCH TRAINING
2022	Yewon Moon, M.S. Incoming graduate student	Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea	Regulatory T-cell suppression assays and scRNA-seq data analysis using R and machine-learning models.
2022	Karoliina Eliisa Ruhno, M.S. Incoming graduate student	Senior Researcher; IMBiologics, Seoul, Republic of Korea	Regulatory T-cell suppression assays.
2023	Moonki Chae, M.S. Student intern	Ph.D. Student; Department of Pathology; Johns Hopkins University, Baltimore, MD, USA	Immune-cell culture, flow cytometry, and scRNA-seq data analysis using machine-learning models and AI tools.
2023	Sung Bin Bae, M.S. Incoming graduate student	Staff, Biologics CMC Team, Yuhan Corporation, Seoul, Republic of Korea	Cell culture, flow cytometry, and GeoMx DSP spatial transcriptomics data analysis.
2024	Mi Jeong Oh, M.S. Manager, PB Immune Therapeutics	Manager; PB Immune Therapeutics, Seoul, Republic of Korea	Experimental design for animal sample collection, ELISA and SPR assay design, and 10× Genomics scRNA-seq.
2024	Eun Sol Jo, B.S. Assistant Manager, PB Immune Therapeutics	Assistant Manager; PB Immune Therapeutics, Seoul, Republic of Korea	Animal experiment design and evaluation of a mouse model of experimental autoimmune encephalomyelitis (EAE).

■ REFERENCES

Chung-Gyu Park, M.D., Ph.D.

Professor, Microbiology & Immunology /
Biomedical Sciences, SNU College of
Medicine; CEO/CSO, PB Immune Therapeutics
· chgpark@snu.ac.kr

Doctoral advisor and industry supervisor; directed the anti-CD40 antibody program.

Hyun Je Kim, M.D., Ph.D.

Assistant Professor, Microbiology &
Immunology / Biomedical Sciences;
Interdisciplinary Program in Artificial
Intelligence (IPAI), SNU · tte9801@snu.ac.kr

Corresponding on the several single-cell studies; supervised the computational work.

Seung-Kuy Cha, Ph.D.

Professor, Dept. of Physiology & Global
Medical Science, Yonsei University Wonju
College of Medicine · skcha@yonsei.ac.kr
Most recent research supervisor (2025–2026).

Jeong Seok Lee, M.D., Ph.D.

Associate Professor, Graduate School of
Medical Science & Engineering, KAIST, Daejeon
· chemami@kaist.ac.kr
Corresponding author, PDE4D / Nature Immunology study.